# ELEG 3143 Assignment # 2

1. Let $E, F, G$ be three events. Find expressions for the events that of $E, F, G$.

   (a) Only $F$ occurs.

   (b) At least two events occur.

   (c) At most two events occur.

2. A card is drawn at random from a standard deck of 52 cards. Let $A$ be the event that a king is drawn, $B$ the event that a spade is drawn, and $C$ the event that a ten of spades is drawn. Describe each of the events listed below and calculate its probability.

   (a) $A \cup B$

   (b) $B \cup C$

   (c) $A \cap C$

   (d) $A \cap B \cap C$

3. An experiment consists of ranomly drawing three cards in succession without replacement from a standard deck of 52 cards. Let $A$ be the event of a king on the first draw, $B$ the event of a king on the second draw, and $C$ the even of a king on the third draw. Describe each of the events listed below and calculate its probability.

   (a) $A \cap B^c$

   (b) $A^c \cap B^c$

   (c) $A^c \cap B^c \cap C^c$

4. Tossing two dice. What is the probability that the first die is six given that the sum of the dice is seven?

5. A family has two children. What is the conditional probability that both are boys given that at least one of them is a boy? Assume that boy and girl are equally likely.

6. You have a shuffled deck of three cards: 2, 3, and 4. You draw one card. Let $C_i$ denote the event that card $i$ is picked, for $i = 2, 3$, or 4. Let $E$ denote the event that the card is an even-numbered card.

   (a) What is $\Pr(C_2|E)$?

   (b) What is $\Pr(E|C_2)$?

7. When a pair of dice are rolled, let $A$ be the event of obtaining a number of 6 or greater and let $B$ be the event of obtaining a number of 6 or less. Are events $A$ and $B$ independt or dependent?